

MSDP SOFTWARE GUIDE

Some remarks about *corrections after calibration* of MSDP channels
Pierre Mein

We present some additional remarks to « MSDP-geom-calib-25-10 » in the case of calibration function, which enables the detection of possible adjustments and corrections of the spectrograph optics.

Calibration function

A **calibration function** $cal(X,Y,n)$ is defined as the flat field intensity divided by the average profile intensity in the same wavelength (taking into account the wavelength function along the Y-coordinate).

Calibrated intensities will be obtained by dividing the observed intensities by the calibration function in each point X,Y,n (n =channel). **If the files of flat fields provide excellent statistics and results**, some little intensity error coming from spectrograph adjustments will be **corrected**.

Optical corrections using calibration function of intensities

Figures 1 to 3 show the calibration function and results obtained for the 9 X*Y channels with the good statistical result derived from the 25 flat field files. Many little disturbances are corrected by intensity calibration.

Their positions and shapes in the channels can also be used to localize and realize new optical adjustments, for example by observing pupil geometry for a given point of a given channel in the output of spectrograph.

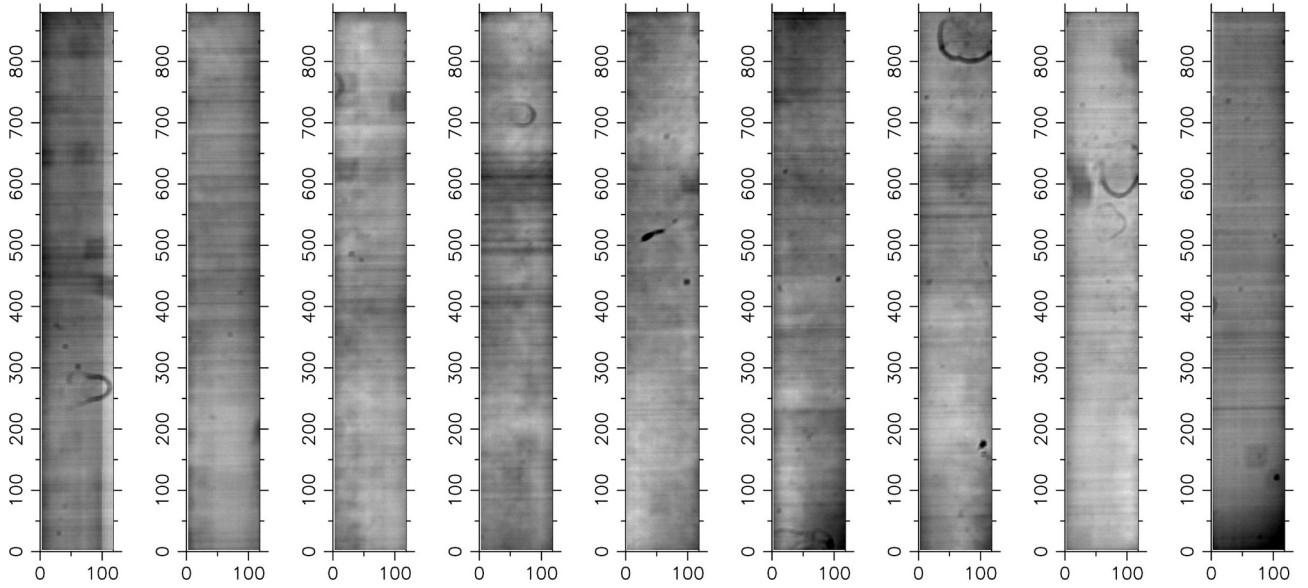


Fig. 1. Calibration function (*cal.ps*)

Fig. 1 shows the calibration function obtained from flat field files between 09:59:39 and 10:01:35. Black and white correspond to values 0.8 and 1.2.

Fig. 2 shows the intensities in a target observation (at 10:20:33, after flat field) without any correction (*obsD1.ps*), divided by average intensity along each X-value (which corresponds almost to constant wavelength). This division allows to show simultaneously all fluctuations and solar structures in all wavelengths of the line profile. Black and white correspond also to 0.8 and 1.2 of this average value.

In both figures we can note some disturbances, such as for example :

- **dark dusts** (channel $n=1$, $y=270$; channel $n=7$, $y=800$; channel $n=8$, $y=600$)
- **pupils reduced** progressively along X and Y (channels $n=6, 7$ and 9 , $y<150$)
- **constant darkness** across one full channel (channel $n=1$, $Y>810$)

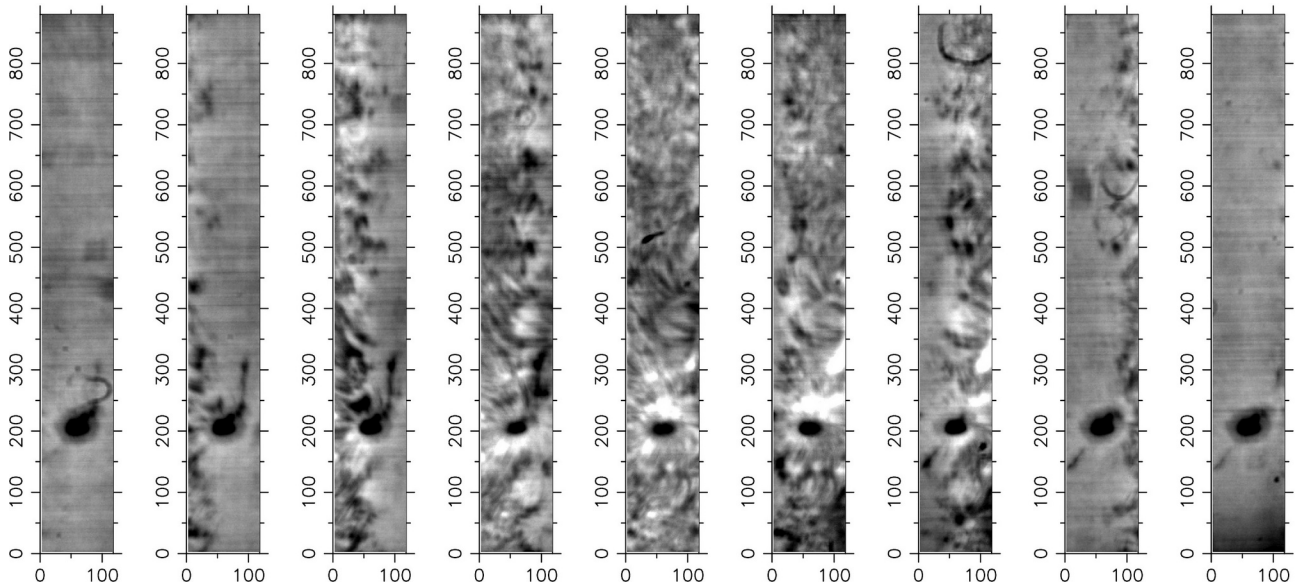


Fig. 2. Observed intensity without calibration (*obsD1.ps*), divided by average intensity along each X value.

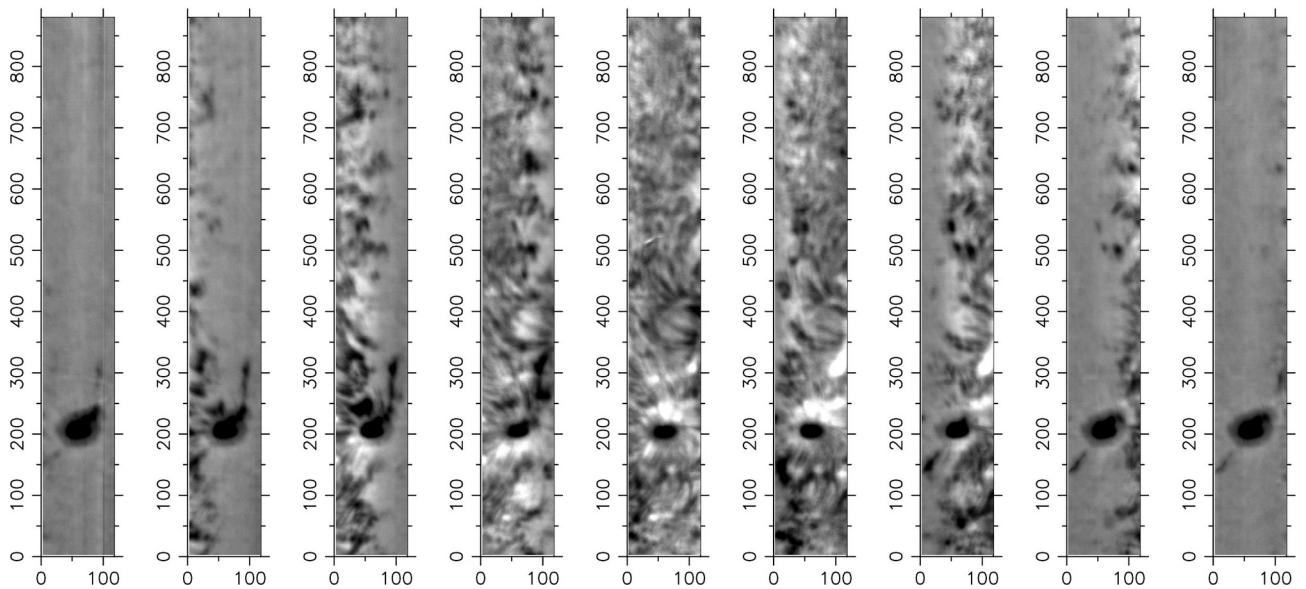


Fig. 3 Same observed intensities, with calibration correction (*obsD2.ps*)

We can see that disturbances noted in Fig. 1 and 2 almost completely disappeared in Fig. 3, proving the efficiency of calibration. Let us remind also that strong solar structures can hide small calibration fluctuations. In the previous examples many corrections can be checked :

- **dark dusts** (channel $n=1$, $y=270$; channel $n=7$, $y=800$; channel $n=8$, $y=600$) **disappeared** ; in channel 7, $y=700$, some solar structures seem to replace the right part of the dust. Probably due to dust on field lens at the output.
- **pupils reduced** along X and Y (channels $n=6, 7$ and 9 , $y<150$) are **now constant** along X and Y.
- **constant darkness** across one full channel (channel $n=1$, $Y>810$) **disappeared**. Maybe full channel corresponds to problems near slicer slits.

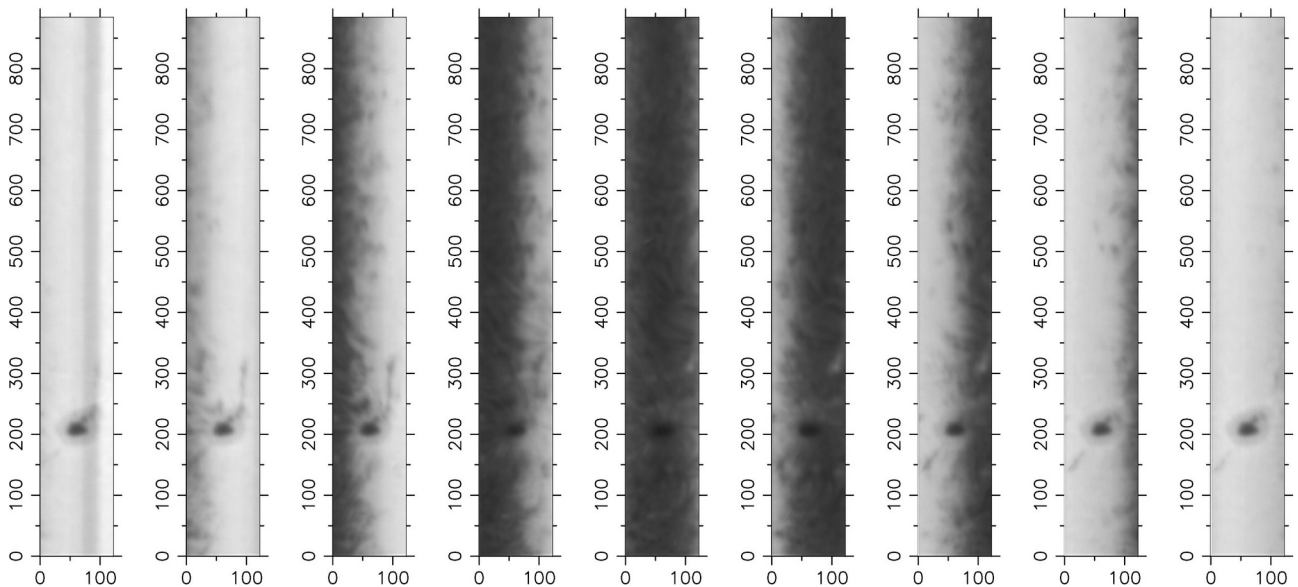


Fig. 4 Observed and calibrated intensity without division by mean value for each X (*obs.ps*).

Fig. 4 is the same as Fig. 3, without division by the average value for each X. Black is close to the intensity of line center, and white is close to the continuum intensity. Disturbances are corrected by calibration, but the sensitivity due to large distance between black and white is much too large to allow useful detections.